

How to Update the Firmware of a MAX32664 on a Custom PCB

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Introduction

Although the [MAX32664 User Guide \(https://www.maximintegrated.com/en/design/technical-documents/app-notes/6/6806.html\)](https://www.maximintegrated.com/en/design/technical-documents/app-notes/6/6806.html) outlines the necessary I2C commands and sequences to update the IC to the latest firmware, implementing this from scratch isn't exactly straightforward. A solution typically involves 3 major components:

- A microcontroller and associated firmware to communicate with the MAX32664 over I2C and GPIOs
- Host PC software/scripts for reading .msbl file data and driving the microcontroller firmware in the right transaction sequence
- A serial API so that the host script and microcontroller can communicate

Luckily, Maxim provides a ready-made solution built around our MAX32630FTHR that can be used as a utility and as a reference implementation.

This guide will walk you through updating the firmware of a MAX32664 on a custom PCB using Maxim's MAX32630FTHR_msbl_flasher tool (https://github.com/MaximIntegratedTechSupport/MAX32630FTHR_msbl_flasher).

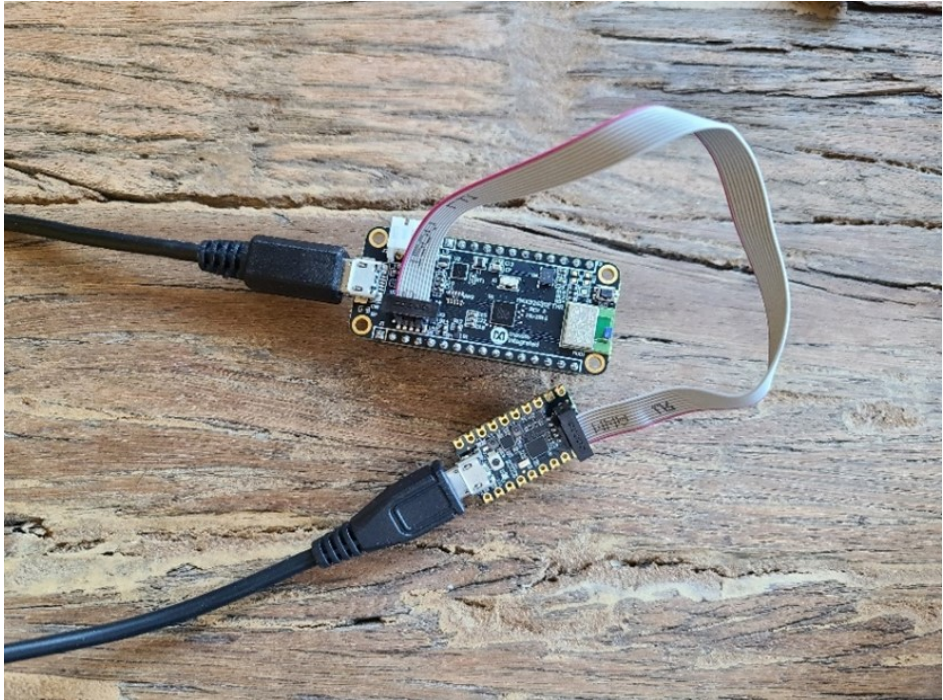
Required Equipment

- [MAX32630FTHR \(https://www.maximintegrated.com/en/products/microcontrollers/MAX32630FTHR.html\)](https://www.maximintegrated.com/en/products/microcontrollers/MAX32630FTHR.html) + [MAX32625PICO \(https://www.maximintegrated.com/en/products/microcontrollers/MAX32625PICO.html\)](https://www.maximintegrated.com/en/products/microcontrollers/MAX32625PICO.html), w/ JTAG Cable (**The MAX32625PICO and cable are included in the MAX32630FTHR package**)
- Your custom PCB with the SLAVE_SDA, SLAVE_SCL, MFIO, and RSTN pins of the MAX32664 exposed.
- The latest .msbl firmware file for your MAX32664 (<https://www.maximintegrated.com/MAX32664.html>) variant, available [here \(https://www.maximintegrated.com/en/products/interface/signal-integrity/MAX32664.html/design-development/tb-tabs-4\)](https://www.maximintegrated.com/en/products/interface/signal-integrity/MAX32664.html/design-development/tb-tabs-4).
- The latest release of MAX32630FTHR_msbl_flasher tool, available [here \(https://github.com/MaximIntegratedTechSupport/MAX32630FTHR_msbl_flasher/releases\)](https://github.com/MaximIntegratedTechSupport/MAX32630FTHR_msbl_flasher/releases).

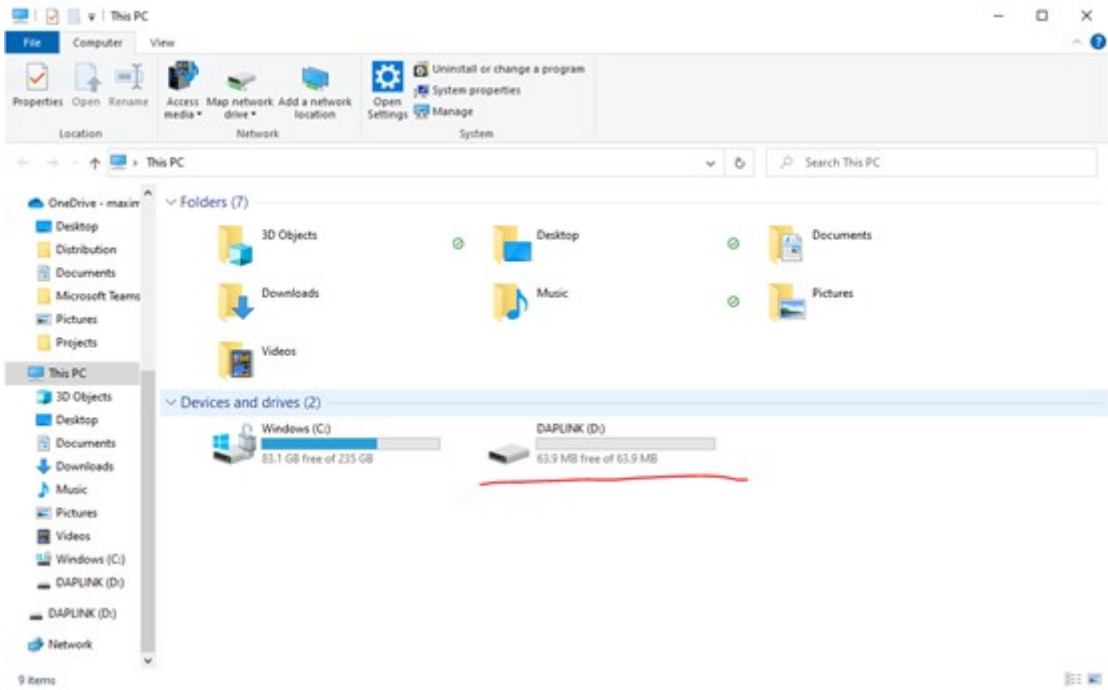
Step 1: Set up the MAX32630FTHR as an msbl programmer

First, we will need to flash a firmware image to the MAX32630FTHR microcontroller. To do so, follow the steps below:

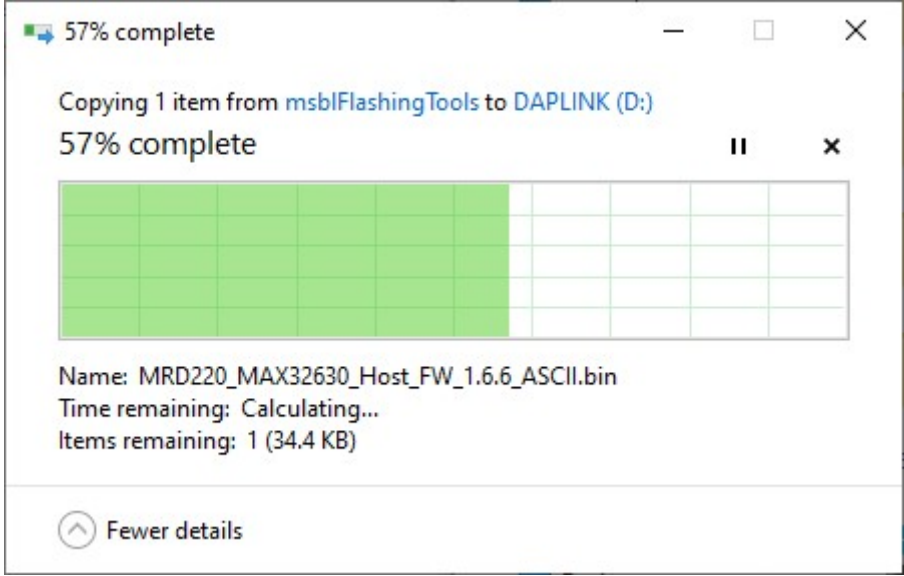
1. Connect the JTAG Cable from the MAX32625PICO to the MAX32630FTHR, and connect *both* devices to your PC with a micro-USB cable.



2. The DAPLINK drag-and-drop programming interface of the MAX32625PICO will appear in your list of devices and drives.



3. Drag and drop the “MAX32630FTHR_msbl_flasher.bin” file from the latest MAX32630FTHR_msbl_flasher software package (https://github.com/MaximIntegratedTechSupport/MAX32630FTHR_msbl_flasher/releases) onto the DAPLINK drive. You will see the LED on the MAX32625PICO blinking as it programs the MAX32630FTHR.



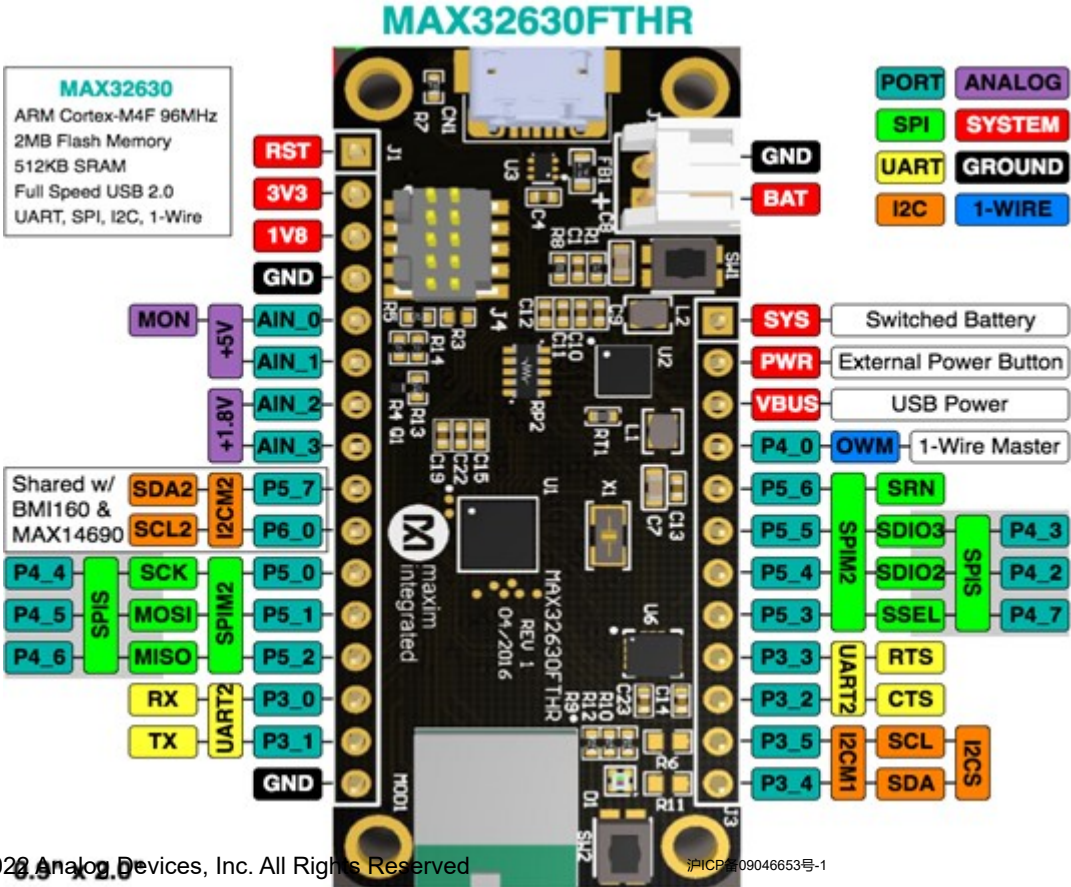
4. Once complete, unplug both devices from your PC. You can remove the JTAG cable and the smaller "PICO" programmer board, it's no longer needed. Your MAX32630FTHR is now ready to be connected to your PCB to flash your MAX32664!

Step 2: Connect the MAX32630FTHR to the MAX32664

Your PCB will need to have the SLAVE_SCL, SLAVE_SDA, MFIO, and RSTN, and VSS (GND) pins of the MAX32664 exposed. Refer to the [MAX32664 Datasheet](https://datasheets.maximintegrated.com/en/ds/MAX32664.pdf) (<https://datasheets.maximintegrated.com/en/ds/MAX32664.pdf>) for pin #s for your package type.

1. Connect the following pins together.

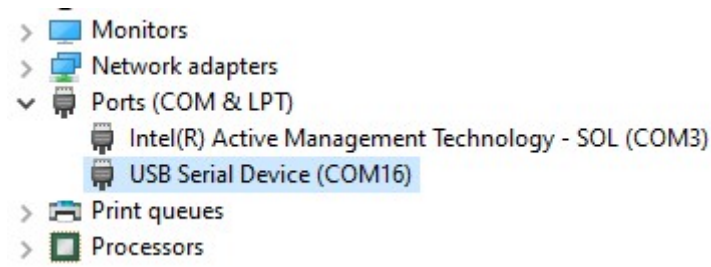
MAX32630FTHR			MAX32664	
Pin #	Description		Pin #	Description
P3_5	I2CM1 SCL	->	See Datasheet	SLAVE_SCL
P3_4	I2CM1 SDA	->	See Datasheet	SLAVE_SDA
P5_6	Used as GPIO	->	See Datasheet	RSTN
P5_4	Used as GPIO	->	See Datasheet	MFIO



2. Connect one of the GND pins of the MAX32630FTHR to the GND reference on your PCB that is connected to V_{SS} (Digital ground) of the MAX32664. **This is very important to ensure I2C and GPIOs work properly.**
3. Provide the necessary 3.3V to V_{DD} on the MAX32664 to power on the IC. This will depend on how your PCB is laid out. The 3.3V supply can come from the 3.3V output on the MAX32630FTHR if necessary. 3.3V is used to match the logic level of the MAX32630FTHR firmware. If your PCB uses a 1.8V supply, a logic level translator should be used for the I2C bus and GPIOs so as not to exceed that absolute max ratings of the MAX32664 pins. It is recommended to use a 3.3V supply for the MAX32664 during the flash for the sake of simplicity.

✓ Step 3: Reconnect the MAX32630FTHR to your PC

Use the micro-USB connector to re-connect the MAX32630FTHR to your PC. **Take note of which COM port the MAX32630FTHR is using.** On Windows, it will show up as a generic USB Serial Device.



✓ Step 4: Update the MAX32664 Firmware

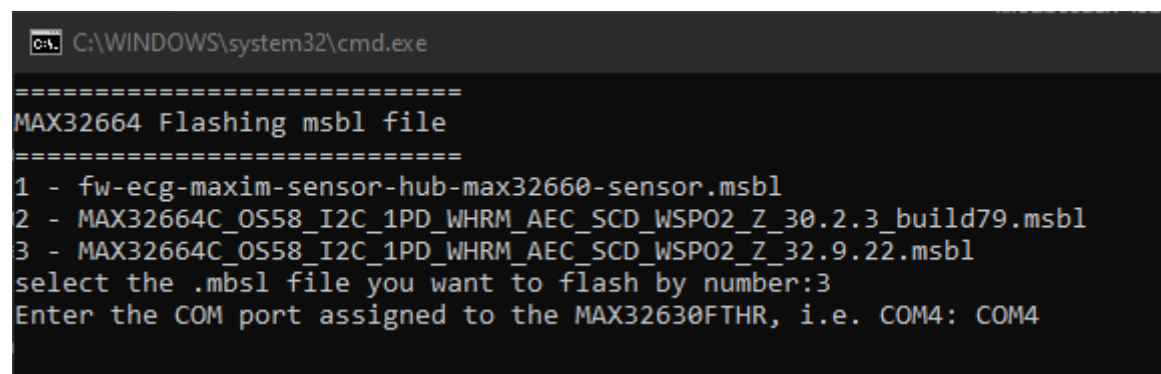
1. (Optional) Copy the .msbl file for your MAX32664 variant into the same directory as the "flash.exe" executable for your own convenience. Otherwise, the -f option below will need to include the full file-path to the .msbl file.
2. Open a command prompt or shell in the same directory as the "flash.exe" executable. Run the executable. You can copy the following command as a template, or run "flash.exe -h" for help.

```
flash.exe -f "[your msbl filename]" -p "[MAX32630FTHR COM port]"
```

If you are using Linux or Mac, the Python scripts inside of the "Python" directory of the software download can be used as an alternative. From within the Python folder...

```
pip install -r requirements.txt
python flash.py -f "[your msbl filename]" -p "[MAX32630FTHR COM port]"
```

Alternatively, on Windows the included "run.bat" file can be used. This file will prompt the user to select from a list of .msbl files that were copied in the same location as the "flash.exe" file. Once the file has been selected by the assigned list number, then the console prompts the user to enter the assigned COM port to the MAX32630FTHR. Once this has been entered the program proceeds to flash the msbl file as shown below:



3. Monitor the command line output for any errors. An example of a successful flash of a MAX32664 on a [MAXREFDES220 PCB](https://www.maximintegrated.com/content/dam/files/secured/design-tools/software/6657/maxrefdes220-schematic.pdf) (<https://www.maximintegrated.com/content/dam/files/secured/design-tools/software/6657/maxrefdes220-schematic.pdf>) is shown below...

```
Command Prompt
Microsoft Windows [Version 10.0.19042.1288]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Jake.Carter>cd C:\Users\Jake.Carter\codespace\MAX32630FTHR_msbl_flasher_v101

C:\Users\Jake.Carter\codespace\MAX32630FTHR_msbl_flasher_v101>flash.exe -f "MAX32664C_HSP2_WHRM_AEC_SCD_WSP02_Z_30.13.19.msbl" -p "COM16"
Reading msbl file MAX32664C_HSP2_WHRM_AEC_SCD_WSP02_Z_30.13.19.msbl...Success!
  Magic: msbl
  Format Version: 0
  Target: MAX32660
  Encoding Type: AES-192
  Num Pages: 29
  Page Size: 8192
  CRC Size: 4
  Size of header: 76
  Resv0: 0
  CRC32: 0xbfeecb84
Connecting to MAX32630FTHR on COM16...Success!
Resetting to application mode...
Sleeping for 3s to let Sensor Hub initialize.....Done!
  Attempting to retrieve version number...Success!
  Current firmware is v30.12.0
Entering bootloader mode...Success!
Getting bootloader version...Success!
  Bootloader version: 3.1.6
Getting page size...Success!
Page size in bootloader matches page size of msbl file.
Setting number of pages to flash from msbl file (29)...Success!
Setting Initialization Vector (IV) bytes...Success!
Setting Authentication bytes...Success!
Erasing existing msbl file...Success!
Flashing msbl file...
  Flashing page 1/29...Success!
  Flashing page 2/29...Success!
  Flashing page 3/29...Success!
  Flashing page 4/29...Success!
  Flashing page 5/29...Success!
  Flashing page 6/29...Success!
  Flashing page 7/29...Success!
  Flashing page 8/29...Success!
  Flashing page 9/29...Success!
  Flashing page 10/29...Success!
  Flashing page 11/29...Success!
  Flashing page 12/29...Success!
  Flashing page 13/29...Success!
  Flashing page 14/29...Success!
  Flashing page 15/29...Success!
  Flashing page 16/29...Success!
  Flashing page 17/29...Success!
  Flashing page 18/29...Success!
  Flashing page 19/29...Success!
  Flashing page 20/29...Success!
  Flashing page 21/29...Success!
  Flashing page 22/29...Success!
  Flashing page 23/29...Success!
  Flashing page 24/29...Success!
  Flashing page 25/29...Success!
  Flashing page 26/29...Success!
  Flashing page 27/29...Success!
  Flashing page 28/29...Success!
  Flashing page 29/29...Success!
Exiting bootloader mode...Success!
Sleeping for 3s to let Sensor Hub initialize.....Done!
Retrieving operating mode...Success!
Retrieving new firmware version...Success!
New firmware version is v30.13.19

C:\Users\Jake.Carter\codespace\MAX32630FTHR_msbl_flasher_v101>
```

Step 5: Verify the Version # of the Updated Firmware

After a successful flash, the host script will attempt to reset the MAX32664 to application mode and read off the new firmware version #. Check the last line of the flash to ensure that the retrieved version # matches. For example...

```
Exiting bootloader mode...Success!
Sleeping for 3s to let Sensor Hub initialize.....Done!
Retrieving operating mode...Success!
Retrieving new firmware version...Success!
New firmware version is v30.13.19
```

Keywords: How To, Update, Firmware, Flash, MAX32664, MAX32664A, MAX32664B, MAX32664C, MAX32664D, Custom, PCB, msbl program, programmer,

